Angle information from the measured tracking error

2026-10-15 x–z plane, $\theta_x = 0$

1 Measurement coordinate and local coordinate

local → measurement: $\delta\bar{\mathbf{p}} = V\,[\delta\bar{u} \;\; \delta\bar{v} \;\; \delta\bar{w}]^{\mathsf{T}}$

$$R_x(\theta_x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_x & -\sin\theta_x \\ 0 & \sin\theta_x & \cos\theta_x \end{bmatrix}$$

$$R_y(\theta_y) = \begin{bmatrix} \cos\theta_y & 0 & \sin\theta_y \\ 0 & 1 & 0 \\ -\sin\theta_y & 0 & \cos\theta_y \end{bmatrix}$$

$$V = R_y(\theta_y)\,R_x(\theta_x) = [\,\mathbf{u} \;\; \mathbf{v} \;\; \mathbf{w}\,]$$

$$\theta_x = 0: \quad \delta\bar{x} = \cos\theta_y\,\delta\bar{u} + \sin\theta_y\,\delta\bar{w}$$

$$\delta\bar{z} = -\sin\theta_y\,\delta\bar{u} + \cos\theta_y\,\delta\bar{w}$$

2 Deterministic and random components of the measured tracking error

$$\delta\bar{\mathbf{p}}_m[k] = \bar{\mathbf{p}}_d[k-2] - \bar{\mathbf{p}}_m[k]$$

$$\overline{\delta\bar{\mathbf{p}}_m}[k+1] = a_{\mathrm{pd}}\,\delta\bar{\mathbf{p}}_m[k] + (1-a_{\mathrm{pd}})\,\overline{\delta\bar{\mathbf{p}}_m}[k]$$

$$\delta\bar{\mathbf{p}}_r[k] = \delta\bar{\mathbf{p}}_m[k] - \overline{\delta\bar{\mathbf{p}}_m}[k] = [\,\delta\bar{x}_r \;\; \delta\bar{y}_r \;\; \delta\bar{z}_r\,]^{\mathsf{T}}$$

3 Variance and covariance in the local coordinate

assumption: the random components along the wall and along the surface normal are uncorrelated

$$\mathrm{E}\big[\delta\bar{u}_r^2\big] = C_T\,\kappa_T\,\bar{a}_T + C_n\,\sigma_n^2$$

$$\mathrm{E}\big[\delta\bar{w}_r^2\big] = C_T\,\kappa_T\,\bar{a}_N + C_n\,\sigma_n^2$$

$$\mathrm{E}\big[\delta\bar{u}_r\,\delta\bar{w}_r\big] = 0$$

4 Local coordinate of the estimated angle

the estimator uses $\hat{\theta}_y$; the wall is at θ_y

$$
\begin{aligned}
e_\theta &= \theta_y - \hat{\theta}_y \\
[\,\delta\tilde{u}_r \;\; \delta\tilde{w}_r\,]^\mathsf{T} &= R_y(\hat{\theta}_y)^\mathsf{T}\,[\,\delta\bar{x}_r \;\; \delta\bar{z}_r\,]^\mathsf{T} \\
&= R_y(\hat{\theta}_y)^\mathsf{T}\,R_y(\theta_y)\,[\,\delta\bar{u}_r \;\; \delta\bar{w}_r\,]^\mathsf{T} = R_y(e_\theta)\,[\,\delta\bar{u}_r \;\; \delta\bar{w}_r\,]^\mathsf{T} \\
\delta\tilde{u}_r &= \cos e_\theta\,\delta\bar{u}_r + \sin e_\theta\,\delta\bar{w}_r \\
\delta\tilde{w}_r &= -\sin e_\theta\,\delta\bar{u}_r + \cos e_\theta\,\delta\bar{w}_r
\end{aligned}
$$

5 Variances and covariance in the estimated local coordinate

$$
\begin{aligned}
\delta\tilde{u}_r^2 &= \cos^2 e_\theta\,\delta\bar{u}_r^2 + 2\sin e_\theta\cos e_\theta\,\delta\bar{u}_r\delta\bar{w}_r + \sin^2 e_\theta\,\delta\bar{w}_r^2 \\
\delta\tilde{w}_r^2 &= \sin^2 e_\theta\,\delta\bar{u}_r^2 - 2\sin e_\theta\cos e_\theta\,\delta\bar{u}_r\delta\bar{w}_r + \cos^2 e_\theta\,\delta\bar{w}_r^2 \\
\delta\tilde{u}_r\,\delta\tilde{w}_r &= -\sin e_\theta\cos e_\theta\,\delta\bar{u}_r^2 + (\cos^2 e_\theta - \sin^2 e_\theta)\,\delta\bar{u}_r\delta\bar{w}_r + \sin e_\theta\cos e_\theta\,\delta\bar{w}_r^2 \\
\\
\mathrm{E}\big[\delta\tilde{u}_r^2\big] &= C_T\,\kappa_T\,(\cos^2 e_\theta\,\bar{a}_T + \sin^2 e_\theta\,\bar{a}_N) + C_n\,\sigma_n^2 \\
\mathrm{E}\big[\delta\tilde{w}_r^2\big] &= C_T\,\kappa_T\,(\sin^2 e_\theta\,\bar{a}_T + \cos^2 e_\theta\,\bar{a}_N) + C_n\,\sigma_n^2 \\
\mathrm{E}\big[\delta\tilde{u}_r\,\delta\tilde{w}_r\big] &= C_T\,\kappa_T\,(-\sin e_\theta\cos e_\theta\,\bar{a}_T + \sin e_\theta\cos e_\theta\,\bar{a}_N) \\
&= -C_T\,\kappa_T\,\sin e_\theta\cos e_\theta\,(\bar{a}_T - \bar{a}_N)
\end{aligned}
$$

6 Estimation error of the angle

$\sin e_\theta \approx e_\theta$, $\cos e_\theta \approx 1$

$$
\begin{aligned}
\mathrm{E}\big[\delta\tilde{u}_r^2\big] &\approx C_T\,\kappa_T\,\bar{a}_T + C_n\,\sigma_n^2 \\
\mathrm{E}\big[\delta\tilde{w}_r^2\big] &\approx C_T\,\kappa_T\,\bar{a}_N + C_n\,\sigma_n^2 \\
\mathrm{E}\big[\delta\tilde{u}_r\,\delta\tilde{w}_r\big] &\approx -C_T\,\kappa_T\,(\bar{a}_T - \bar{a}_N)\,e_\theta \\
e_\theta &\approx -\frac{\mathrm{E}\big[\delta\tilde{u}_r\,\delta\tilde{w}_r\big]}{C_T\,\kappa_T\,(\bar{a}_T - \bar{a}_N)}
\end{aligned}
$$

7 Measured input gains and measured angle

one sample in place of $\mathrm{E}[\,\cdot\,]$; estimates in place of $\bar{a}_T$, $\bar{a}_N$

$$\bar{a}_{Tm}[k] = \frac{\delta\tilde{u}_r^2[k] - C_n\,\sigma_n^2}{C_T\,\kappa_T}$$
$$\bar{a}_{Nm}[k] = \frac{\delta\tilde{w}_r^2[k] - C_n\,\sigma_n^2}{C_T\,\kappa_T}$$
$$e_{\theta m}[k] = -\frac{\delta\tilde{u}_r[k]\,\delta\tilde{w}_r[k]}{C_T\,\kappa_T\left(\hat{\bar{a}}_T[k] - \hat{\bar{a}}_N[k]\right)}$$
$$\theta_{ym}[k] = \hat{\theta}_y[k] + e_{\theta m}[k]$$

expectations without the approximation (Section 5), random component as in Section 3 (not changed by $\hat{\theta}_y$ through the control law)

$$\mathrm{E}\big[\bar{a}_{Tm}\big] = \cos^2 e_\theta\,\bar{a}_T + \sin^2 e_\theta\,\bar{a}_N$$
$$\mathrm{E}\big[\bar{a}_{Nm}\big] = \sin^2 e_\theta\,\bar{a}_T + \cos^2 e_\theta\,\bar{a}_N$$
$$\mathrm{E}\big[e_{\theta m}\big] = \frac{\bar{a}_T - \bar{a}_N}{\hat{\bar{a}}_T - \hat{\bar{a}}_N}\,\sin e_\theta \cos e_\theta$$

Fig. 1 Measured angle and measured input gains with the estimated angle held fixed

simulation: plane wall $\theta_y = 20°$; command along the surface normal to $\bar{h} = 1.2$, then held 5 s; motion gains known in the control law; $e_\theta = 0°, \pm 1°, \pm 2°, \pm 5°$ (small angle errors, the assumption of Section 6), $\hat{\theta}_y$ not updated; one sample per step (Section 6), mean over the hold, paired by seed (each seed minus its own reading at $e_\theta = 0$) ± SEM, 15 seeds; $\lambda_c = 0.4$; red = expectations of Section 6 with $\bar{a}_T$, $\bar{a}_N$ at the hold (as Fig. 2); left: $e_{\theta m}$; right: $\bar{a}_{Nm}/\bar{a}_N$ (red solid) and $\bar{a}_{Tm}/\bar{a}_T$ (red dashed)

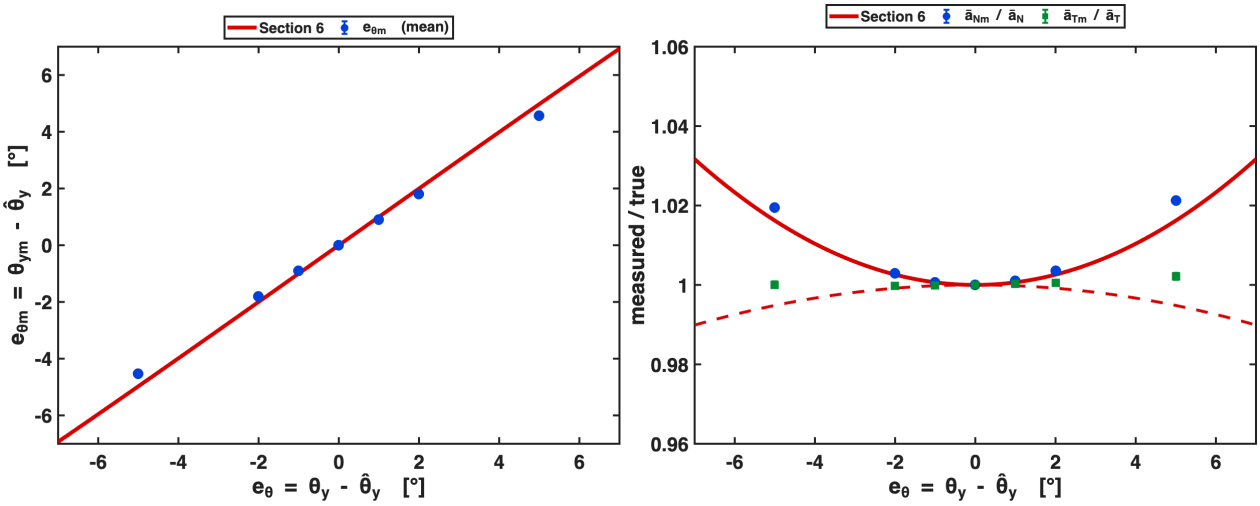


Fig. 2 Section 5 checked: control law with the wall angle, readout in the local coordinate of $\hat{\theta}_y$

simulation as Fig. 1, but the control law uses $\theta_y = 20°$; the same random component read with $e_\theta = 0°, \pm 1°, \pm 2°, \pm 5°$ (small angle errors); mean over the hold, paired by seed (each seed minus its own reading at $e_\theta = 0$) ± SEM, 15 seeds; red = expectations of Section 6 (from Section 5) with $\bar{a}_T$, $\bar{a}_N$ at the hold; left: $\bar{a}_{Nm}/\bar{a}_N$; middle: $\bar{a}_{Tm}/\bar{a}_T$; right: $e_{\theta m}$

